Optimal strategies in the all-heads coin game

Verified in Lean 4, written with Claude Opus 4.7

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The game

- n coins; each flip: H with prob. p, T with prob. $q = 1 - p$
- each round: flip the in-play coins; keep ≥ 1 head H
- kept coins move into the right column and carry over
- bottom row all green $\Rightarrow$ *win*; no head $\Rightarrow$ *lose*

Strategy One

R1 H T H T

Strategy All

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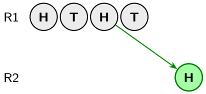
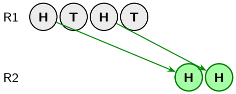


Strategy All



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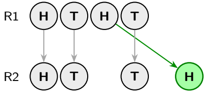

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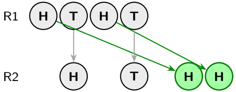

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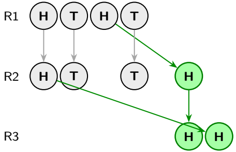

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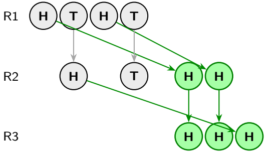

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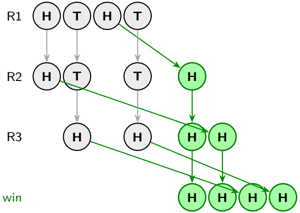

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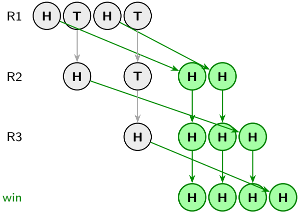


The Bellman equation

$w_{n,p}$ = optimal winning probability with n coins, $w_{0,p} := 1$.

$$\boxed{w_{n,p} \;=\; p^n \;+\; \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \le m \le n-1} w_{m,p}}$$

- p^n: all heads $\Rightarrow$ immediate win
- j tails: choose remaining $m \in \{j, \ldots, n-1\}$
- player maximises $w_{m,p}$

The fair case $p = \frac{1}{2}$

Theorem

$w_{n,1/2} = \frac{1}{2}$ for every $n \geq 1$.

More: every policy taking the immediate win at $k = n$ has $w_{n,1/2} = \frac{1}{2}$.

Proof. Strong induction on n; key identity $\sum_{j=1}^{n-1} \binom{n}{j} 2^{-n} = 1 - 2^{1-n}$:

$$w_{n,1/2} \;=\; 2^{-n} + \tfrac{1}{2}\left(1 - 2^{1-n}\right) \;=\; \tfrac{1}{2}. \qquad \square$$

- symmetry axis
- all reasonable strategies optimal

Above: $p > \frac{1}{2}$

Theorem

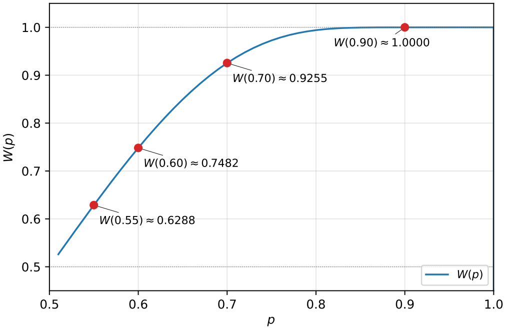
For $p > \frac{1}{2}$, $n \geq 2$:

(i) $w_{n,p} > w_{n-1,p}$

(ii) ONE optimal: $w_{n,p} = a_{n,p}$

$\Rightarrow W(p) := \lim_n w_{n,p}$ exists,
$p \leq W(p) < 1$,

$$W(p) = \sum_{k\geq 1} p^k \prod_{j>k} (1 - p^j - q^j).$$



Proof idea: joint induction with $w_{n-1,p} < \dfrac{p^n}{p^n + q^n}$.

Workflow: human + Claude Opus 4.7

Manuscript, code, Lean formalisation: produced *interactively*.

Author

- question: $w_{n,p}$ vs. $v_{n,p}$
- perturbation in $\delta = \frac{1}{2} - p$
- joint induction
- decision to formalise
- review every step

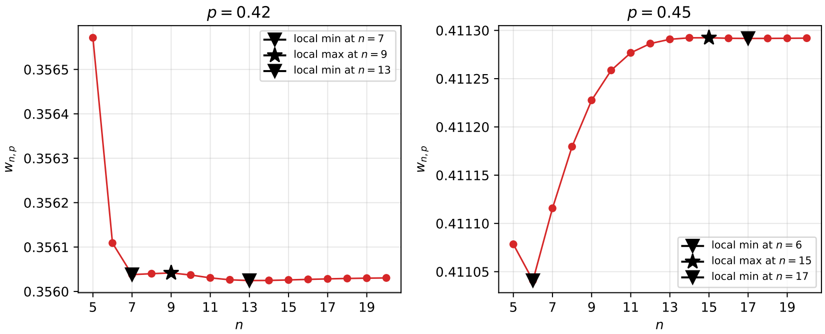
Claude

- drafts prose
- proposes / debugs Lean tactics
- picks Mathlib lemmas
- spots cubic / Tannery
- numerics, plots

Loop: pick result $\to$ draft $\to$ lake build $\to$ review $\to$ commit.

Log: journal.md $\cdot$ ≈ 3700 lines Lean $\cdot$ 0 sorry $\cdot$ 0 custom axioms.

Below: $p < \frac{1}{2}$ is hard



- local minima & maxima drift with p
- neither ONE nor ALL optimal in general
- our route: perturbation around $p = \frac{1}{2}$, $p = \frac{1}{2} - \delta$

First-order asymptotics: c_n

Deficit $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$, $p = \frac{1}{2} - \delta$:

$$\Delta_{n,p} = c_n\,\delta + O(\delta^2), \qquad c_1 = 1,$$

$$\boxed{c_n = \frac{n}{2^{n-1}} + \frac{1}{2^n}\sum_{j=1}^{n-1}\binom{n}{j}\min_{j\le m\le n-1} c_m} \quad (n \ge 2).$$

$c_1 = 1, \quad c_2 = \frac{3}{2}$

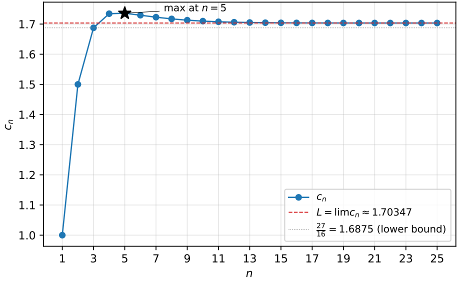
$c_3 = \frac{27}{16}$

$c_4 = \frac{111}{64}$

$c_5 = \frac{3555}{2048}$ (max)

$c_6 = \frac{113337}{65536}$

$c_n \to L \approx 1.7035$



Shape of $w_{n,p}$ at first order

Corollary ($p = \frac{1}{2} - \delta$, $\delta > 0$ small)

(i) $w_{n-1,p} - w_{n,p} = (c_n - c_{n-1})\,\delta + O(\delta^2)$

(ii) strict local *minimum* at $n = 5$

(iii) no local maximum at first order

Proof idea. Joint induction on n over four statements:

(a) *collapse:* $\min_{j \leq m \leq n-1} c_m = \begin{cases} c_j & j \leq 3 \\ c_{n-1} & j \geq 4 \end{cases} \qquad (n \geq 7)$

(b) *lower bound:* $c_n \geq \frac{27}{16} \qquad (n \geq 4)$

(c) *strict decrease:* $c_n < c_{n-1} \qquad (n \geq 6)$

(d) *linear recursion:* $c_n = A_n + (1 - B_n)\, c_{n-1} \qquad (n \geq 7)$

Lean 4: Challenge.lean

Trust surface = Challenge.lean + Defs.lean.

Comparator: only propext, Quot.sound, Classical.choice.

```
-- Defs.lean : Bellman definition of w
noncomputable def w (p : ℝ) : ℕ → ℝ
  | 0     => 1
  | n + 1 =>
    p ^ (n + 1)
      + ∑ j ∈ (Ico 1 (n + 1)).attach,
        (Nat.choose (n + 1) j.val : ℝ)
          * p ^ (n + 1 - j.val) * (1 - p) ^ j.val
          * (Ico j.val (n + 1)).attach.sup' _ (fun m => w p m.val)

-- Challenge.lean : one of 16 headline statements
theorem w_local_min_at_five :
  ∃ δ₀ : ℝ, 0 < δ₀ ∧ ∀ δ, 0 < δ → δ < δ₀ →
    w (1/2 - δ) 5 < w (1/2 - δ) 4 ∧
    w (1/2 - δ) 5 < w (1/2 - δ) 6
```

16 theorems · 0 sorry · 0 custom axioms · github.com/pfaffelh/coins